

27 A by Heisenberg's uncertainty principle,

$$\Delta p \Delta x \geq h$$

$$\Delta p = \left(\frac{0.20}{100} \right) m_e v$$

$$\begin{aligned} \Delta x &= \frac{h}{\Delta p} = \frac{h}{\left(\frac{0.20}{100} \right) m_e v} \\ &= \frac{6.63 \times 10^{-34}}{\left(\frac{0.20}{100} \right) (9.11 \times 10^{-31}) (1.5 \times 10^6)} \\ &= 2.4 \times 10^{-7} \text{ m} \end{aligned}$$

28 $C = C_0 - (C_0) \exp(-\lambda t) + C_0$